Comparative Study of Automated Brain Tumor Segmentation Techniques.

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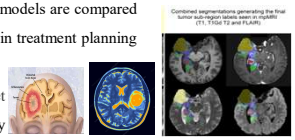
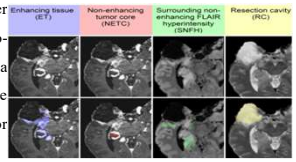
Problem Statement

A Comparative Study For Automated Segmentation Models To Improve Tumor Region Delineation In MRI Images Of Diffuse Gliomas

Diffuse gliomas(stages I-III) are hard to treat as they are infiltrative. Various automated segmentation models are compared in this study to determine the best approach for accurately defining tumor regions in MRI images to aid in treatment planning and outcome prediction.



DATA: We are using the BRATS2025 dataset (Brain Tumor Segmentation) which is a benchmark dataset for evaluating AI models in brain tumor segmentation from MRI scans. Scans consisted of natively acquired T1, post-contrast T1-weighted (T1Gd), T2-weighted (T2), and T2 FLAIR volumes obtained over different clinical protocols and institutions. Pre-processing pipelines guaranteed equivalency through co-registration, 1mm³ interpolation, and skull stripping. Multiple raters have manually annotated image data according to a set protocol that was neuroradiologist-approved. Annotations include growing tumor tissue (ET - label 3), non-enhancing FLAIR hyperintensity around it (SNFH - label 2), non-enhancing tumor core (NETC - label 1), and resection cavity (RC - label 4), with the RC label being a recent addition.

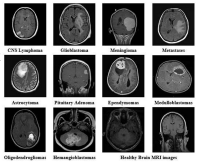


Overall Approach

Step 1: Problem Definition

•Identify the segmentation task

•Define evaluation metrics



Step 2: Data Preparation

•Collect and preprocess data (normalization, augmentation, 2D/3D patches)

•Split into training, validation, and test sets.

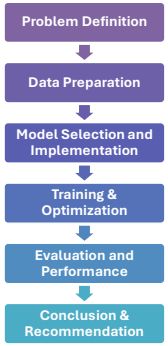
Step 3: Model Selection And Implementation

•Implement Inception U-Net (multi-scale feature extraction).

•Implement ResUNet (residual blocks for deeper networks).

•Implement Attention U-Net (attention gates for feature focusing)

•Implement 3D U-Net (volumetric data processing)



Step 4: Training & Optimization

•Use consistent hyperparameters (learning rate, batch size, loss function).

•Apply cross-validation for robustness.

•Optimize using techniques like early stopping, LR scheduling

Step 6: Evaluation & Performance

•Compare performance on test data (segmentation accuracy, speed).

•Analyze computational efficiency (training time, memory usage).

•Assess model strengths/weaknesses for different data types

Step 7: Conclusion And Recommendations

•Determine the best model based on task requirements.

•Suggest hybrid or modified architectures if needed

Solution

CHALLENGES:

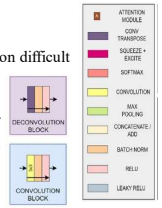
1.Infiltrative Nature of Diffuse Gliomas – Tumor boundaries are unclear, making segmentation difficult

2.Variability in MRI Scans – Different imaging protocols and scanners affect consistency.

3.Preprocessing Complexity – Requires interpolation, and skull stripping for standardization.

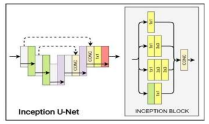
4.Annotation Variability – Manual annotations by multiple raters may introduce subjectivity.

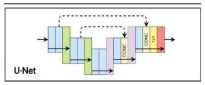
5.Model Performance Evaluation – Identifying the most accurate segmentation model.

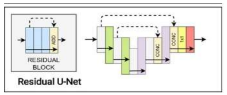


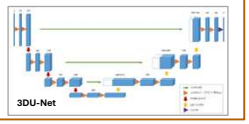
SOLUTION:

Our approach evaluates several advanced deep learning architectures for segmentation: Inception U-Net, ResUNet, Attention U-Net, and 3D U-Net. Inception U-Net excels in handling varying object sizes through multi-scale feature extraction. ResUNet offers improved gradient flow with residual connections, promoting stability in deeper networks. Attention U-Net uses attention gates to prioritize relevant features, enhancing efficiency. Lastly, 3D U-Net adapts U-Net for volumetric data, making it suitable for medical imaging. The choice among them depends on specific task needs, such as multi-scale processing, training stability, feature focusing, or 3D data handling.









Milestones

Model Implementation (Week 1-2)

Using the provided MRI datasets, the team will implement and establish baseline models for the classification task. During this step, we will pick and implement the task-appropriate model architectures. Preprocessing pipelines will also be developed to ensure the data is properly prepared and ready for training.

Model Training (Week 3)

After implementing the models, we will train them. This phase will involve setting up the training loop, loss functions, and optimization algorithms. We will run the models on the training dataset for a predetermined amount of epochs and monitor their performance to ensure that they are learning efficiently. The training results will be recorded and analyzed.

Fine-Tuning (Week 4)

Once the models have been trained, we will look into fine-tuning techniques to improve their performance. This will entail experimenting with hyperparameters, attempting transfer learning with pre-trained models, modifying learning rates, and assessing various regularization procedures to prevent overfitting. Fine-tuning may also include rewriting certain layers or adding data augmentation techniques.

Validation and Performance reporting (Week 5)

After fine-tuning, we'll test the models on the validation set to see how well they generalize to new data. Key performance measures (e.g., accuracy, F1-score, Dice score) will be calculated, and a report will be prepared summarizing the models' performance. We will also look into potential faults and correct any anomalies in predictions.